BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Subprograms and Arrays

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. When might a procedure be suitable?

2. In OCR ERL, s = 'VARIABLE'. What is s.substring(1,3)?

A) VAR
B) RIA
C) ARI
D) RIAB

3. "Failure of any one workstation cable necessarily disables all other star-network links." Correct this. Explain why.

4. A network must cope if one link fails. "A mesh has only one path between any two devices." Correct this. How could it keep working?

Today's objectives

Organise larger programs into subprograms. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

Subprogram
A procedure or function defines reusable instructions.

Array parameter
An array can be supplied to a subprogram so values can be processed.

Return value
A function passes a result back to the code that called it.

Testing
Test a subprogram using several arrays including boundary cases.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Subprogram	A procedure or function defines reusable instructions. In the score-array helper, identify where this technique applies.
Array parameter	An array can be supplied to a subprogram so values can be processed. In the library stock function, identify where this technique applies.
Return value	A function passes a result back to the code that called it. In the sports result processor, identify where this technique applies.
Testing	Test a subprogram using several arrays including boundary cases. In the classroom rating program, identify where this technique applies.
Passing and returning an array	A function may receive an array as an argument, process it and return an array result. For example, receive [2,4,6], create a new array by adding 1 to each element, then return [3,5,7]. The caller must use the returned result. Passing the input and returning the output are separate actions; the original changes only if the implementation changes it.

I Do - One worked example

A score-array helper uses subprogram and array parameter. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Subprogram [1]; A procedure or function defines reusable instructions. [1].

3. Array parameter [1]; An array can be supplied to a subprogram so values can be processed. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A library stock function needs subprogram. Explain the purpose. [2 marks]

2. Explain array parameter for a sports result processor. [2 marks]

3. When programming a classroom rating program, explain why return value and testing matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Finish the comparison that lets the function return the largest array value.

```
function largest(values)
biggest = values[0]
for i = 1 to 2
if ... then
biggest = values[i]
endif
next i
return biggest
endfunction
array disks = [32,128,64]
print(largest(disks))
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain array parameter and return value in a temperature array report. [4 marks]

Q4. A programmer makes a competition leaderboard. Explain the roles of subprogram, return value and testing. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A procedure or function defines reusable instructions. ☐ T ☐ F
2. A procedure or function defines reusable instructions ☐ T ☐ F
3. An array can be supplied to a subprogram so values can be processed ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 For a named reusable task such as displaying a menu when no function result is required. ☐

Q2 C - ARI ☐

Q3 If the central switch fails, devices using it lose contact. A broken cable usually affects just one workstation. ☐

Q4 A mesh has other paths if some links fail. More links add cost and complexity. Check the paths that remain. ☐

Code First - Model

```
function largest(values)
biggest = values[0]
for i = 1 to 2
if values[i] > biggest then
biggest = values[i]
endif
next i
return biggest
endfunction
array disks = [32,128,64]
print(largest(disks))
```

Output: 128

We Do - Check the model

1. Subprogram [1]; A procedure or function defines reusable instructions. [1]. ☐

2. Array parameter [1]; An array can be supplied to a subprogram so values can be processed. [1]. ☐

3. Return value [1]; A function passes a result back to the code that called it. [1]. Testing [1]; Test a subprogram using several arrays including boundary cases. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Subprogram [1]; Array parameter [1]. ☐

Q3 Array parameter [1]; An array can be supplied to a subprogram so values can be processed. [1]. Return value [1]; A function passes a result back to the code that called it. [1]. ☐

Q4 Subprogram [1]; A procedure or function defines reusable instructions. [1]. Return value [1]; A function passes a result back to the code that called it. [1]. Testing [1]; Test a subprogram using several arrays including boundary cases. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: